Student Notes and Lab Exercises

Chapter 1 — What the Brain Is (and Is Not)

This companion document is designed to reinforce the conceptual foundations established in Chapter 1. The emphasis here is not memorization, but reflection, comparison, and systems-level reasoning.

Key Learning Objectives

- Identify why computational metaphors can mislead when applied to biological intelligence.
- Explain the brain as a physical, biological system rather than an abstract information processor.
- Describe the brain as an adaptive control system operating under evolutionary constraints.
- Articulate why noise, redundancy, and apparent inefficiency are functional features of neural systems.

Conceptual Summary

Chapter 1 reframes the brain as an evolved, adaptive system optimized for survival rather than correctness. Key themes include physical constraint, historical contingency, and the functional role of noise and redundancy.

Lab 1: Brains vs Computers (Conceptual Comparison)

In this lab, you will construct an explicit comparison between a digital computer and a biological brain, focusing on architecture, constraints, and failure modes.

Tasks

- Create a two-column table comparing brains and digital computers across at least five dimensions.
- Identify one feature that appears inefficient in the brain but advantageous from a biological perspective.
- Write a short reflection on how this comparison changes your intuition about intelligence.

Lab 2: No Reset Button

This lab explores the implications of irreversible history in biological systems.

Tasks

- Describe how the absence of a reset function affects learning and habit formation.
- Give a real-world example where early experiences constrain later behavior.
- Compare this with a software system that allows full reinitialization.

Optional Extension: AI Reflection

Reflect on how modern machine learning systems align with or diverge from biological principles discussed in Chapter 1.